Sampling & Estimation

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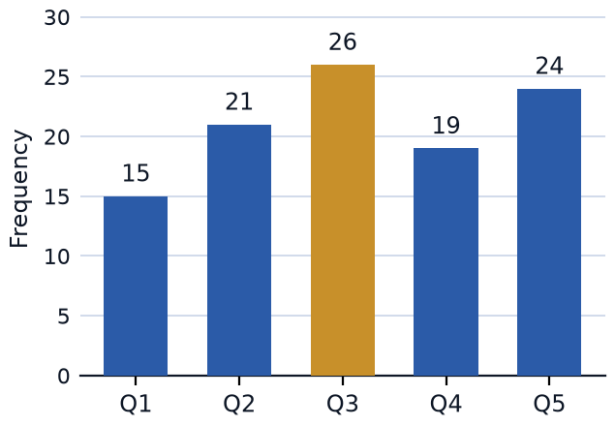
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

A random sample of 10 values of a variable x is taken. The results are summarised as follows: $\sum x = 50, \quad \sum x^2 = 322$ Calculate an unbiased estimate of the population variance.

2 [Foundational] • Target: The Central Limit Theorem (CLT)

A random sample of $n = 50$ observations is taken from a population. The sample values are summarised as follows: $\sum x = 450, \quad \sum x^2 = 4295.$ Find an unbiased estimate of the population variance.



3 [Foundational] • Target: Confidence Intervals for the Population Mean

The random variable X has a Poisson distribution with mean $\lambda > 0$. Given that $3\,\mathrm{P}(X = 2) = \mathrm{P}(X = 4)$, find the value of λ.

4 [Foundational] • Target: Confidence Intervals for Large Unknown-Variance Samples

A curve satisfies the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{k - y}{2x + 1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

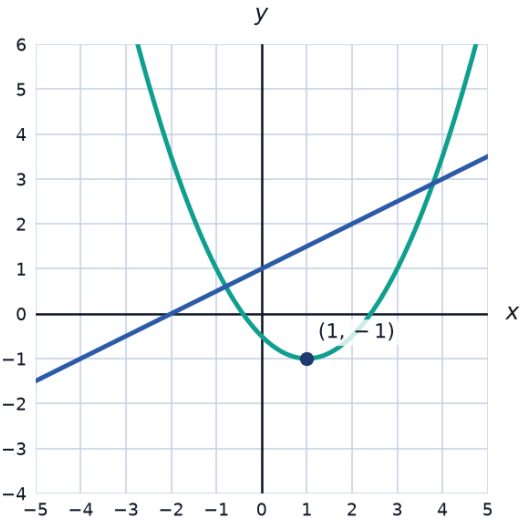
5 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

Solve the equation $\ln(3x + 1) - 2\ln 2 = \ln(x - 1)$.

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6 [Foundational] • Target: The Central Limit Theorem (CLT)

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.



7 [Foundational] • Target: Confidence Intervals for the Population Mean

The continuous function f is defined for all $x \in \mathbb{R}$ by $f(x) = 2^x + 3x - 11$. Between which pair of consecutive integers does a root of the equation $f(x) = 0$ lie?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

The first three terms of a geometric progression are given by $u_1 = 3k - 2$, $u_2 = k + 2$, $u_3 = k - 2$, where k is a real constant. Given that the infinite geometric series $\sum_{n=1}^{\infty} u_n$ converges, find the sum to infinity, S_∞.

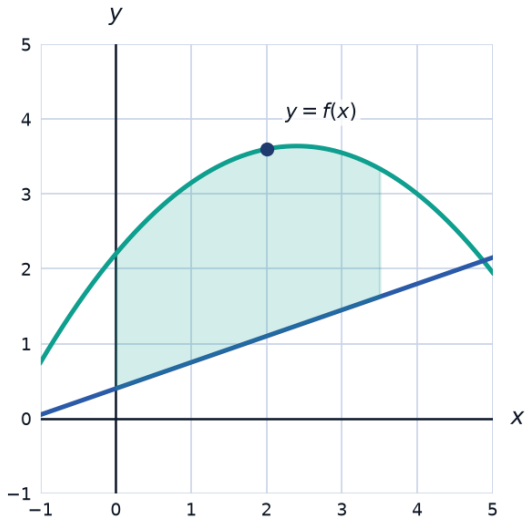
9 [Intermediate] • Target: Unbiased Estimators of Population Mean & Variance

Relative to a fixed origin O, the position vectors of the points A and B are given by $\vec{OA} = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$. Find the unit vector in the direction of $\vec{AB}$.

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10 [Intermediate] • Target: The Central Limit Theorem (CLT)

Find the exact value of the definite integral $\int_0^4 \frac{1}{2x+1}\,\mathrm{d}x$.



11 [Intermediate] • Target: Confidence Intervals for the Population Mean

The continuous random variable X has probability density function given by

$f(x) = \{k(4-x^2), 0 \le x \le 2, ; 0, \text{otherwise},$ where k is a constant. Find the value of k.

12 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

Given that $\cot\theta = -\frac{3}{4}$ and $\frac{\pi}{2} < \theta < \pi$, find the exact value of $\operatorname{cosec}\theta$.

13 [Intermediate] • Target: Unbiased Estimators of Population Mean & Variance

Find the exact real solution to the equation $2e^{2x} + 5e^{x} - 12 = 0$.

14 [Intermediate] • Target: The Central Limit Theorem (CLT)

Find the particular solution of the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} = xe^{-y}$ given that $y = 0$ when $x = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

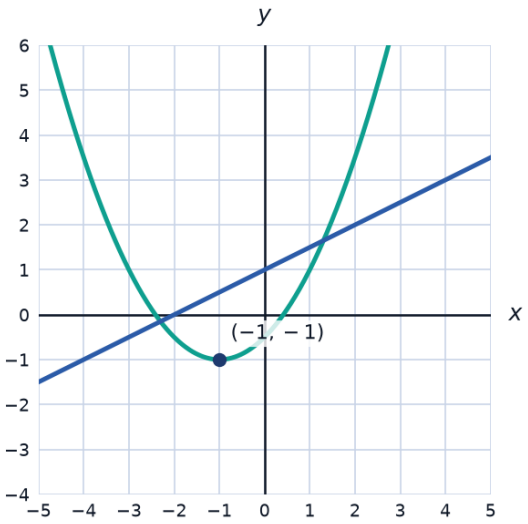
15 [Challenging] • Target: Confidence Intervals for the Population Mean • Cross-Lesson: 15.01 Hypothesis Testing

Find the exact value of the definite integral $\int_0^{\ln 2} e^{2x}\,\mathrm{d}x$.

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16 [Challenging] • Target: Confidence Intervals for Large Unknown-Variance Samples • Cross-Lesson: 2.01 Exponentials & Logarithms

Relative to a fixed origin O, the points A and B have position vectors $\overrightarrow{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\overrightarrow{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\overrightarrow{AB}$.



17 [Challenging] • Target: Unbiased Estimators of Population Mean & Variance • Cross-Lesson: 3.01 Advanced Trigonometry

The complex number z satisfies the equation $z(2 - \mathrm{i}) = 7 + 4\mathrm{i}$. Express z in the form $x + \mathrm{i}y$, where x and y are real.

18 [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 4.01 Differentiation

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

19 [Challenging] • Target: Confidence Intervals for the Population Mean • Cross-Lesson: 5.01 Integration

Find $\int x(x^2 + 1)^5 \, dx$ using substitution.

20 [Challenging] • Target: Confidence Intervals for Large Unknown-Variance Samples • Cross-Lesson: 6.01 Sequences & Series

Evaluate $\int_1^3 (x + 2)^2 \, dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

What is the unbiased estimator of the population variance σ^2 calculated from a random sample of size n?

a. $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$

b. $s^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$

c. $s^2 = \frac{1}{n+1}\sum(x_i - \bar{x})^2$

d. $s^2 = \sum(x_i - \bar{x})^2$

22 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

A random sample of 10 observations has $\sum x = 80$ and $\sum x^2 = 724$. Calculate the unbiased estimate of the population variance.

a. 9.33

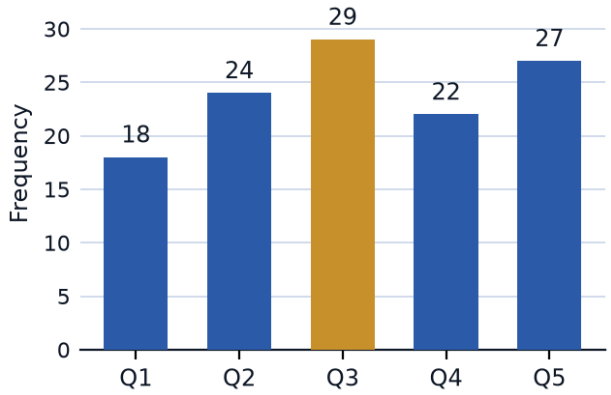
b. 8.40

c. 10.50

d. 7.20

23 [Foundational] • Target: Unbiased Estimators of Population Mean & Variance

If the biased sample variance is $v = 18$ from a sample of size $n = 25$, find the unbiased estimate s^2.



a. 18.75

b. 17.28

c. 18.00

d. 19.50

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24 [Foundational] • Target: The Central Limit Theorem (CLT)

State the Central Limit Theorem for the distribution of the sample mean $\bar{X}$.

a. For large n, $\bar{X}$ is approximately normal $N(\mu, \sigma^2/n)$ regardless of the population distribution

b. For any n, $\bar{X}$ is strictly normal

c. For large n, the population becomes normal

d. $\bar{X}$ has mean $n\mu$ and variance $n\sigma^2$

25 [Foundational] • Target: The Central Limit Theorem (CLT)

A population has mean 100 and standard deviation 15. A sample of size $n = 36$ is taken. What is the standard error of the sample mean?

a. 2.5

b. 15

c. 4.17

d. 0.417

26 [Foundational] • Target: The Central Limit Theorem (CLT)

Using the distribution from the previous question ($\mu = 100$, SE $= 2.5$), find $P(\bar{X} > 105)$.

a. 0.0228

b. 0.0500

c. 0.0456

d. 0.0100

27 [Foundational] • Target: Confidence Intervals for the Population Mean

A sample of size $n = 64$ from a normal population with known $\sigma = 8$ has sample mean $\bar{x} = 52.0$. Find the 95% confidence interval for μ.

a. $[50.04, 53.96]$

b. $[49.50, 54.50]$

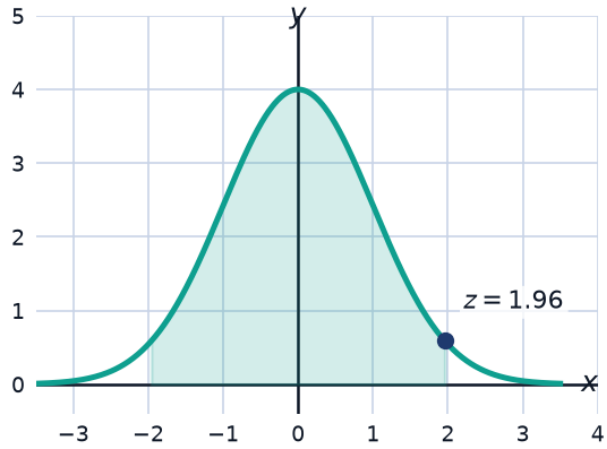
c. $[50.36, 53.64]$

d. $[51.00, 53.00]$

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28 [Foundational] • Target: Confidence Intervals for the Population Mean

What is the critical value z used to construct a 99% confidence interval for the population mean?



a. 2.576

b. 1.960

c. 1.645

d. 2.326

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Confidence Intervals for the Population Mean

How large must a sample be so that the width of a 95% confidence interval for μ is at most 2 units, given $\sigma = 10$?

a. At least 385

b. At least 100

c. At least 196

d. At least 250

30 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

When σ^2 is unknown but the sample size $n = 100$ is large, what is substituted for σ^2 in calculating the confidence interval?

a. The unbiased sample variance estimate s^2

b. The sample range divided by 4

c. The median absolute deviation

d. A default value of 1.0

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31 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

A sample of 100 lightbulbs has mean life $\bar{x} = 1200$ hours and unbiased sample variance $s^2 = 2500$. Find the 90% confidence interval for the true mean life.

a. $[1191.8, 1208.2]$

b. $[1180.0, 1220.0]$

c. $[1190.2, 1209.8]$

d. $[1195.0, 1205.0]$

32 [Intermediate] • Target: Confidence Intervals for Large Unknown-Variance Samples

What is the correct probabilistic interpretation of a 95% confidence interval $[L, U]$ for μ?

a. In repeated sampling, 95% of such constructed intervals will contain the true parameter μ

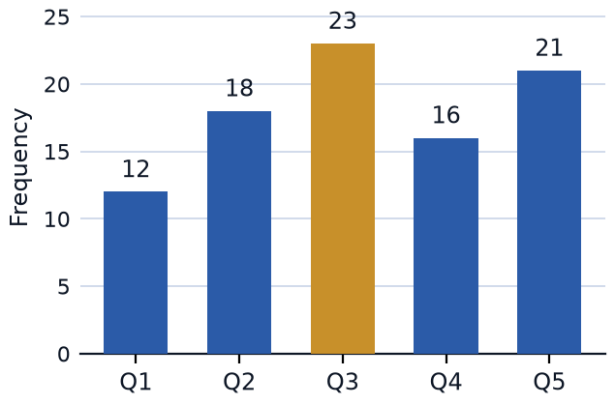
b. There is a 95% probability that μ lies in this specific interval

c. 95% of population values fall between L and U

d. The sample mean falls in $[L, U]$ 95% of the time

33 [Intermediate] • Target: Unbiased Estimators of Population Mean & Variance

A sample of size 10 has $\sum x = 420$ and $\sum x^2 = 19424$. Find the unbiased estimate of the population variance.



a. $\frac{1784}{9}$

b. $\frac{1793}{9}$

c. $\frac{9712}{5}$

d. $\frac{892}{5}$

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34 **[Intermediate]** • **Target: The Central Limit Theorem (CLT)**

A sample of size 10 has $\sum x = 350$ and $\sum x^2 = 13030$. Find the unbiased estimate of the population variance.

a. $\frac{260}{3}$

b. $\frac{263}{3}$

c. 1303

d. 78

35 **[Intermediate]** • **Target: Confidence Intervals for the Population Mean**

A sample of size 8 has $\sum x = 392$ and $\sum x^2 = 19402$. Find the unbiased estimate of the population variance.

a. $\frac{97}{4}$

b. $\frac{9701}{4}$

c. $\frac{194}{7}$

d. $\frac{201}{7}$

36 **[Intermediate]** • **Target: Confidence Intervals for Large Unknown-Variance Samples**

A sample of size 8 has $\sum x = 368$ and $\sum x^2 = 17188$. Find the unbiased estimate of the population variance.

a. $\frac{4297}{2}$

b. $\frac{267}{7}$

c. $\frac{260}{7}$

d. $\frac{65}{2}$

37 **[Intermediate]** • **Target: Unbiased Estimators of Population Mean & Variance**

A sample of size 10 has $\sum x = 120$ and $\sum x^2 = 1672$. Find the unbiased estimate of the population variance.

a. $\frac{116}{5}$

b. $\frac{836}{5}$

c. $\frac{232}{9}$

d. $\frac{241}{9}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 4.01 Differentiation

A sample of size 7 has $\sum x = 322$ and $\sum x^2 = 15500$. Find the unbiased estimate of the population variance.

a. $\frac{15500}{7}$

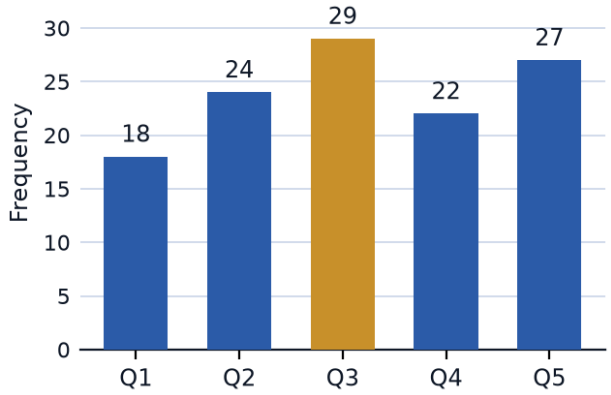
b. $\frac{347}{3}$

c. $\frac{344}{3}$

d. $\frac{688}{7}$

39 [Challenging] • Target: Confidence Intervals for the Population Mean • Cross-Lesson: 5.01 Integration

A sample of size 8 has $\sum x = 376$ and $\sum x^2 = 17978$. Find the unbiased estimate of the population variance.



a. $\frac{153}{4}$

b. $\frac{306}{7}$

c. $\frac{8989}{4}$

d. $\frac{313}{7}$

40 [Challenging] • Target: Confidence Intervals for Large Unknown-Variance Samples • Cross-Lesson: 6.01 Sequences & Series

A sample of size 6 has $\sum x = 282$ and $\sum x^2 = 13312$. Find the unbiased estimate of the population variance.

a. $\frac{6656}{3}$

b. $\frac{58}{5}$

c. $\frac{63}{5}$

d. $\frac{29}{3}$

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41 [Challenging] • Target: Unbiased Estimators of Population Mean & Variance • Cross-Lesson: 7.01 Vectors

A sample of size 6 has $\sum x = 234$ and $\sum x^2 = 9262$. Find the unbiased estimate of the population variance.

a. $\frac{68}{3}$

b. $\frac{4631}{3}$

c. $\frac{141}{5}$

d. $\frac{136}{5}$

42 [Challenging] • Target: The Central Limit Theorem (CLT) • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A sample of size 5 has $\sum x = 230$ and $\sum x^2 = 10726$. Find the unbiased estimate of the population variance.

a. $\frac{146}{5}$

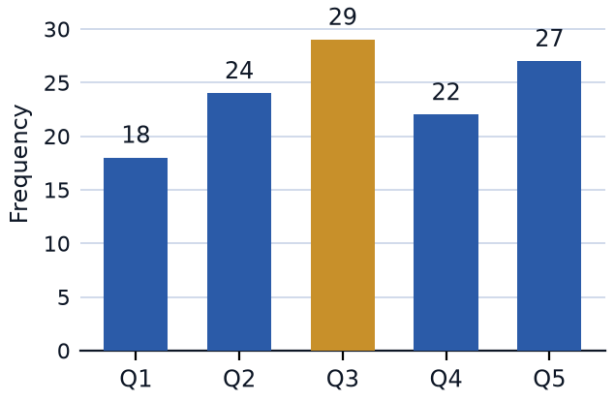
b. $\frac{73}{2}$

c. $\frac{75}{2}$

d. $\frac{10726}{5}$

43 [Challenging] • Target: Confidence Intervals for the Population Mean • Cross-Lesson: 9.01 Complex Numbers

A sample of size 6 has $\sum x = 114$ and $\sum x^2 = 2392$. Find the unbiased estimate of the population variance.



a. $\frac{1196}{3}$

b. $\frac{226}{5}$

c. $\frac{113}{3}$

d. $\frac{231}{5}$

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44 [Challenging] • Target: Confidence Intervals for Large Unknown-Variance Samples • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A sample of size 10 has $\sum x = 220$ and $\sum x^2 = 5098$. Find the unbiased estimate of the population variance.

a. $\frac{86}{3}$

b. $\frac{2549}{5}$

c. $\frac{129}{5}$

d. $\frac{89}{3}$

45 [Challenging] • Target: Unbiased Estimators of Population Mean & Variance • Cross-Lesson: 10.01 Numerical Methods

A sample of size 8 has $\sum x = 176$ and $\sum x^2 = 4680$. Find the unbiased estimate of the population variance.

a. 101

b. $\frac{815}{7}$

c. 585

d. $\frac{808}{7}$

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